

ALEJANDRO DIAZ PAREDES

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SUMMARY

Software engineer building open-source Windows developer tools and full-stack web applications. Hands-on experience with native C/C++ integration, Go installers, Python and TypeScript web stacks, database-backed APIs, and automated testing. Focused on reliable updates, data recovery, and practical tools for users.

RELEVANT SKILLS

- Languages: C/C++, Go, Python, Java, JavaScript, TypeScript, HTML/CSS
- Web and data: Django, Django REST Framework, React, Next.js, PostgreSQL/PostGIS, SQLite
- Systems and tooling: Win32, WebView2, Ghidra, MCP, Git, GitHub Actions
- Cloud and workflows: AWS, Cloudflare, CI/CD, regression testing, REST APIs

EDUCATION

University of Colorado Colorado Springs, May 2026

Colorado Springs, CO

Bachelor of Science in Computer Science, Software Engineering

PROJECTS

Snapmap+ - DOOM 2016 Creator Tools | github.com/doom-snapmap/snapmap-plus

C/C++, Go, Win32, WebView2, JavaScript, HTML/CSS, GitHub Actions

- Reverse-engineered Chrispy's closed-source SnapHak and initiated Snapmap+, an open-source DOOM 2016 editor extension with native hooks and a Win32/WebView2 companion UI.
- Built a Go installer and self-updater with staged executable replacement, rollback paths, and hash-verified cleanup that preserves modified files and existing recovery copies.
- Implemented journaled migrations for user-authored content and older packages, with interrupted-run recovery and checks that protect existing map and resource data.
- Coordinated UI requests with engine-thread execution and timeout handling, distinguishing queued from running operations.
- Added regression tests for interrupted migrations, failed updates, and engine requests that outlive UI timeouts.
- Maintained Windows CI for native, JavaScript, Worker, and Go tests; shipped versioned beta installers and packaged releases for the SnapMap community.

Starview - Personal Astronomy Platform | starview.app | [GitHub](https://github.com)

Django, React, PostgreSQL/PostGIS, TanStack Query, Mapbox, AWS, Cloudflare

- Built and maintained a stargazing platform with location discovery, light-pollution maps, weather and moon conditions, reviews, photos, and saved places.
- Integrated Open-Meteo weather forecasts, PyEphem lunar calculations, and GeoTIFF light-pollution data to provide location-specific observing conditions.
- Implemented TanStack Query caching and optimistic updates with rollback; optimized Django APIs with related-object prefetching, database annotations, and lean list serializers.
- Moved AI-generated review summaries to cached reads and a batch-refresh workflow, keeping external model calls out of interactive page requests.

Reverse-Engineering Developer Tools | [re-discipline](https://github.com/re-discipline) | [GhidraMCP fork](https://github.com/GhidraMCP)

Go, SQLite FTS5, Java, Python, MCP, Ghidra

- Built re-discipline, a Go/SQLite retrieval tool with identifier-aware ranking, source-linked evidence, and regression tests, exposed through CLI, MCP, and HTTP interfaces.
 - Extended LaurieWired/GhidraMCP into a shared headless service with concurrent reads, coordinated writes, revision checks, and cancellation/recovery support.
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WORK EXPERIENCE

Mikron Automation, Controls Engineer Intern, May - Aug 2024

Englewood, CO

- Developed ladder logic automation following Rockwell standards for manufacturing systems.
- Worked with mechanical and electrical engineering teams to verify installed I/O devices.

UCCS, IT Technician I, Sept 2023 - May 2025

Colorado Springs, CO

- Tracked incidents and documented resolutions in the Cherwell ticketing system.
 - Resolved 20+ daily service tickets using Azure, Active Directory, and Microsoft Intune in a university serving 15,000+ users.
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